PRODUCT MANAGEMENT OFFICE

# Second Brain AI

Product Requirements Document (PRD)

**Author:** Product Management Team

**Version:** 1.0 (Draft Blueprint)

**Target Product:** AI-powered Personal Knowledge Management (PKM) SaaS

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# 1. Product Vision & Market Problem

The **Second Brain AI** platform is an advanced Personal Knowledge Management (PKM) software ecosystem designed to empower modern professionals, scholars, and teams. By combining state-of-the-art semantic processing with dynamic indexing models, the system completely removes information friction—allowing users to save, catalog, search, and interrogate their aggregated knowledge repository effortlessly.

> **Vision Statement**
> Second Brain AI acts as an intelligent digital extension of mind. Instead of forcing manual curation, it processes multiple inputs through continuous contextual mapping, enabling natural multi-turn chat interaction and localized discovery.

## 1.1 The Core Problem & Objectives

Information fragmentation has reached a critical threshold. High-value insights remain siloed across incompatible third-party services—including Google Docs, Notion workspaces, detached local PDF whitepapers, email chains, and disconnected chat logs. Standard keyword-based lookups consistently fail because they depend on verbatim phrasing matches, ignoring thematic contexts. Second Brain AI fixes this fragmentation via specialized Retrieval-Augmented Generation (RAG) loops and vector embeddings to drive automated context linking.

# 2. Target Cohorts & User Profiles

The product optimizes workspace functionality around discrete target user segments to deliver targeted value metrics:

### Knowledge Workers & Content Planners

- **Needs:** Track recurring meeting items, extract standard operating procedures, and organize content pipelines.
- **Pain Points:** Managing scattered tracking notes and disconnected browser bookmarks.

### Engineers & Researchers

- **Needs:** Cross-reference documentation libraries, match structural code modules, and inspect large technical whitepapers.
- **Pain Points:** Combing through deeply nested PDF folders.

# 3. Comprehensive Functional Requirements

To meet our core objectives, the system must support the following explicit operational modules:

## 3.1 Core Platform Modules

- **Authentication Guarding:** Multi-tenant security isolation featuring standard email-password signups, OAuth federated configurations, session lifecycle verification, and background tracking audits.
- **Workspace Separation:** Granular administrative borders enabling multi-workspace selection to perfectly separate personal research environments from highly sensitive enterprise projects.
- **Rich Content Editing:** Markdown-compatible canvas tools boasting automated snapshot generation, inline tags, state auto-saving, and real-time validation checks.
- **RAG Document Ingestion:** Native file processing support for complex multi-page PDF documents, docx schemas, raw text, and localized markdown blocks. This pipeline parses and transforms raw inputs via asynchronous text chunkers.
- **Semantic Conversational Interface:** Low-latency conversational loops powered by token-efficient context streaming. Responses include verifiable structural data citations mapping to original files.

# 4. Non-Functional & Operational Guardrails

| Dimension | Target Engineering Thresholds & Compliance Metrics |
|---|---|
| **Performance Efficiency** | Initial structural views render within 2 seconds. Global hybrid semantic search calls return matches in under 1 second. Streaming AI answers initiate context streaming within 3 seconds. |
| **System Scalability** | Multi-tenant data partitioning configured to support high concurrent workloads across millions of distinct vector embedding objects. |
| **Security Enforcement** | End-to-end data transport security utilizing TLS configurations, secure cryptographically-signed cookies, integrated XSS filters, and isolated cross-origin protection. |
| **System Reliability** | Uptime stability target of 99.9%, backed by automated node failures retries and persistent data replica mirroring. |

# 5. Product In-Scope Blueprint & Future Milestones

## 5.1 Minimum Viable Product (MVP) Bounds

The initialization phase will prioritize standalone core capabilities: single-tenant workspace operations, core note-taking text editors, basic asynchronous file loaders (PDF, TXT, DOCX), basic background RAG processing, text similarity searches, and basic streaming chat interfaces. External multi-user collaboration layers, real-time sync systems, mobile apps, OCR data extractors, and billing pipelines are explicitly out of scope for the baseline rollout.

## 5.2 Multi-Phase Strategy Roadmap

- **Phase 2 (Team Cohesion):** Advanced multi-user shared workspace spaces, role assignment matrix mapping, and collaborative feedback fields.
- **Phase 3 (Cognitive Automation):** Automatic categorization engines, text intent tagging, cross-document semantic synthesis, and contextual task mining.
- **Phase 4 (Enterprise Integration):** Enterprise SSO hooks, cloud directory synchronization, compliance tracking dashboards, and customizable data residency parameters.

# 6. Risk Profiling & System Mitigation Strategies

| Identified Risk Vector | Inherent Impact | Strategic Core Mitigation Scheme |
| --- | --- | --- |
| **AI Engine Hallucinations** | High | Strict context alignment enforcement within the RAG extraction chain. Force mandatory citation rendering. |
| **Large File Bottlenecks** | High | Offload expensive processing into asynchronous task managers and use clear file processing progress bars. |
| **High Cloud Compute Overhead** | High | Deploy multi-tier similarity search caching layers and apply usage quotas to the consumption endpoints. |